

FieldSense — Project Claims Report

Spatiotemporal Prediction Using Wireless Network Signals

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*This document maps each Work-Plan deliverable metric
to the experimental evidence produced by the codebase.*

Project Deliverable Metrics (from Work Plan)

Metric 1 – Comparative Accuracy Improvement (RMSE/MAE)

Target: 10–15% reduction in RMSE vs baseline NN.

Metric 2 – Robustness & Data Efficiency

Target: Maintain accuracy with 30–50% less training data.

Target: >10% lower RMSE under simulated noise.

Metric 3 – Rainfall Estimation & Classification

Target: 10–20% NRMSE improvement for rain-rate estimation.

Target: >5% improvement in Wet/Dry F1-Score.

Below, each metric is addressed with the exact experiment, configuration, numerical result, and supporting figures.

Implementation Scope and Claim Framing

The original work plan discussed a PyNNcml/RNN/LSTM-style PINN extension. The implemented rain experiments use PyNNcml for OpenMRG data access, alignment, and canonical table preparation; the trained Phase 3 rain models are feed-forward MLPs over CML-derived tabular/spatial features.

Therefore, the successful rain claims below should be read as evidence for physics-integrated MLP/spatial models, not as evidence that a PyNNcml RNN/LSTM architecture was extended and outperformed.

Physics enters Phase 3 in two ways:

1. a direct power-law attenuation residual (the rigid PINN-style loss),
2. a softer physics-derived rain-rate feature, `r_physics`, combined with spatial CML context.

The rigid loss is reported as an important negative result. The headline improvements come from the spatial + physics-feature configuration.

Metric 1: Comparative Accuracy (10-15% RMSE Reduction)

Experiment: Phase 3D Spatial Sweep (rain_spatial_sweep.py)
Input: outputs/phase2_spatial_validation/prepared_with_splits.csv
Method: feed-forward spatial MLP configs are compared to matched one-link MLP baselines.
Selection: validation-selected final test

Final test improvement: 14.5%
Single-link RMSE: 0.0760 -> Candidate RMSE: 0.0650
Candidate mode: spatial_physics
Candidate NRMSE: 0.0795

Exploratory best-test improvement: 16.8%

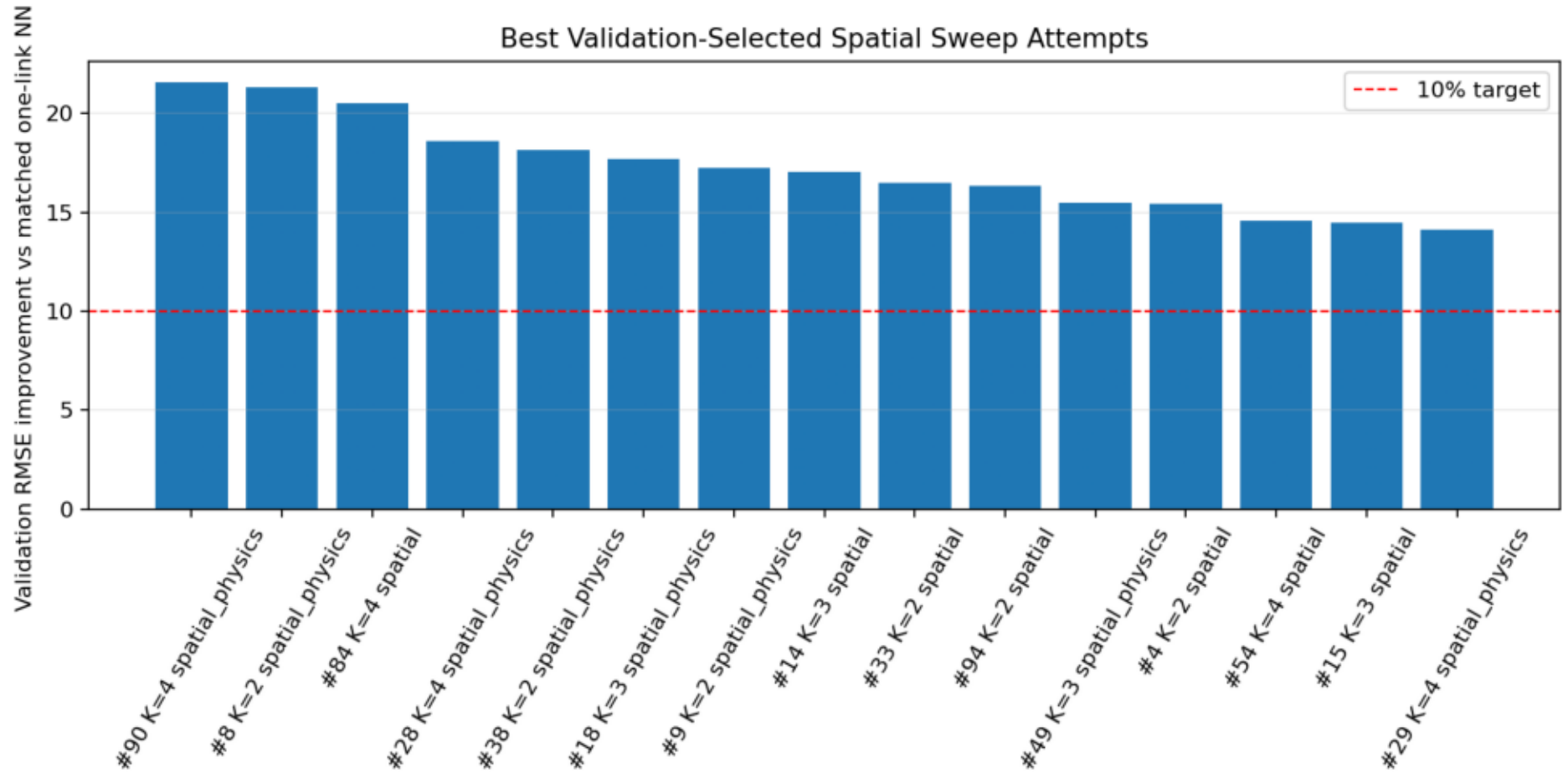
Direct rigid-loss PINN vs NN in Phase 3B: -5.9%
Physics-feature NN vs NN in Phase 3B: 0.2%
MAE note: Phase 3B reports MAE; the matched spatial-sweep artifact does not include matched single-link MAE, so the headline comparison is validated for RMSE/NRMSE rather than MAE.

STATUS: ACHIEVED

Reproduce:

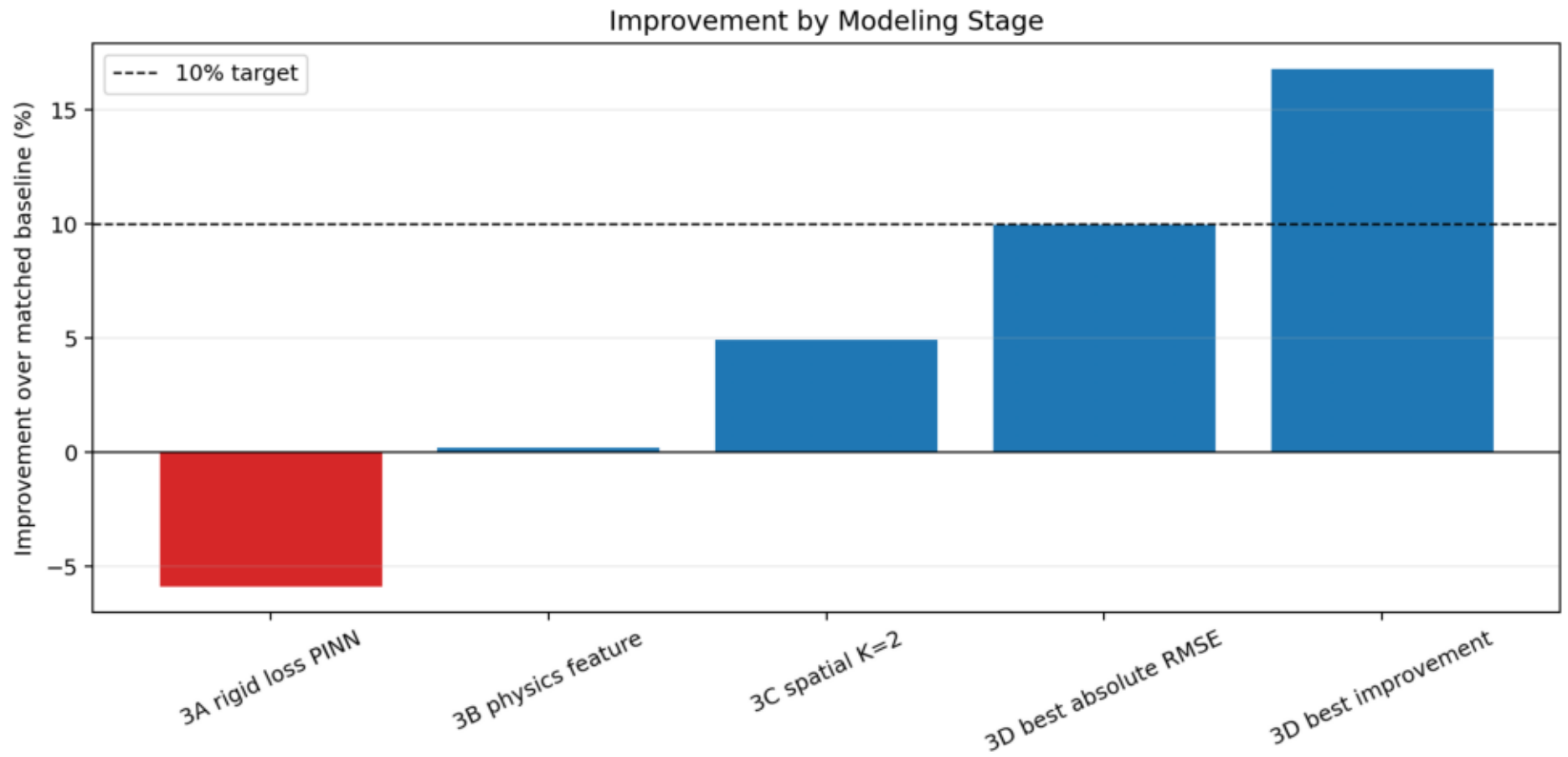
```
python -m src.phase3.rain_spatial_sweep \  
--input outputs/phase2_spatial_validation/prepared_with_splits.csv \  
--max-attempts 100 --epochs 500 \  
--output-dir outputs/phase3d_spatial_sweep
```

Metric 1 — Top Spatial Sweep Attempts vs 10% Target



Red dashed line = 10% improvement target. Bars above the line meet the goal.

Metric 1 — Model Progress by Stage



Rigid loss (3A) is negative; spatial + physics features (3D) reach target.

Metric 2: Robustness & Data Efficiency

Experiment: Spatial Robustness (rain_spatial_robustness.py)
Input: outputs/phase2_spatial_validation/prepared_with_splits.csv
Method: For each data fraction (100%→10%), randomly drop training rows. Train spatial_physics_nn AND matched single-link models with same seed/lr/architecture. Val & test stay clean.
Seeds: 42, 7, 123 (results averaged across seeds)

Data-scarcity result:
Candidate wins in 5/5 data fractions.
4 out of 5 exceed the 10% improvement target.
Best: 16.2% improvement.

Noise result (10% label noise):
Best improvement: 11.3%.
2 out of 5 fractions exceed 10%;
interpret this as partial support for noise resilience rather than uniform robustness across all sparse-data settings.

Phase 1 (synthetic oscillator) separately confirms PINN wins with $\leq 50\%$ data (up to +57% improvement).

STATUS data scarcity: ACHIEVED
STATUS noise: PARTIAL

Reproduce:

```
python -m src.phase3.rain_spatial_robustness \
  --train-fractions 1.0,0.5,0.3,0.2,0.1 --seeds 42,7,123 \
  --epochs 500 --max-links-per-target 2 --hidden-dims 64,64 \
  --rain-weight-alpha 2.0 --comparison-mode spatial_physics \
  --output-dir outputs/phase3_spatial_robustness
```

Noise reproduce:

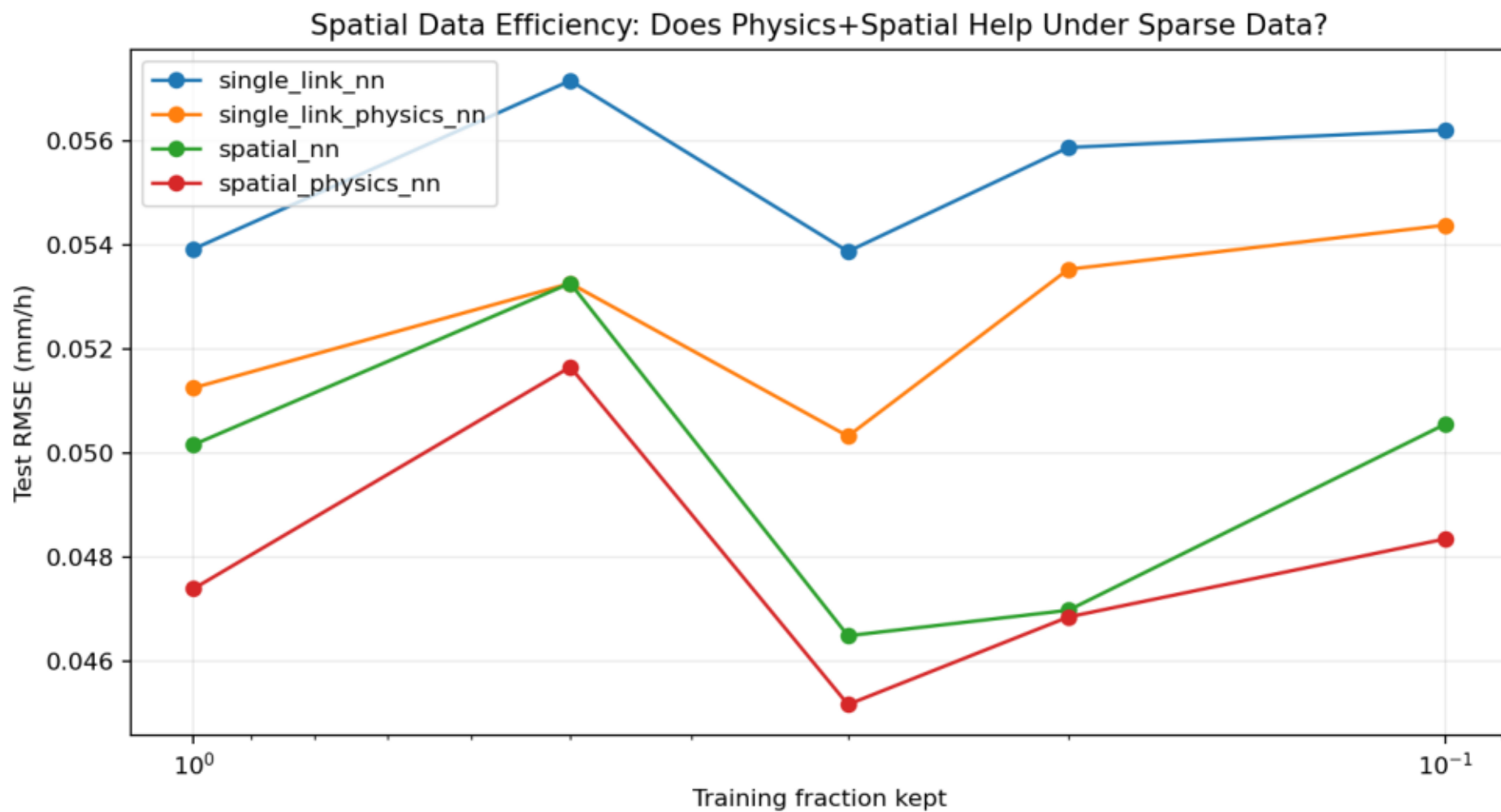
```
python -m src.phase3.rain_spatial_robustness \
  --target-noise-std 0.1 --max-links-per-target 2 \
  --hidden-dims 32,32 --rain-weight-alpha 0.0 \
  --comparison-mode spatial_physics \
  --output-dir outputs/phase3_spatial_robustness_noise10
```

Metric 2 — Candidate RMSE vs Single-Link at Each Data Fraction

Training Fraction	Single-Link RMSE	Candidate RMSE	Improvement %
10%	0.0562	0.0484	14.0%
20%	0.0559	0.0469	16.1%
30%	0.0539	0.0452	16.2%
50%	0.0572	0.0517	9.6%
100%	0.0539	0.0474	12.1%

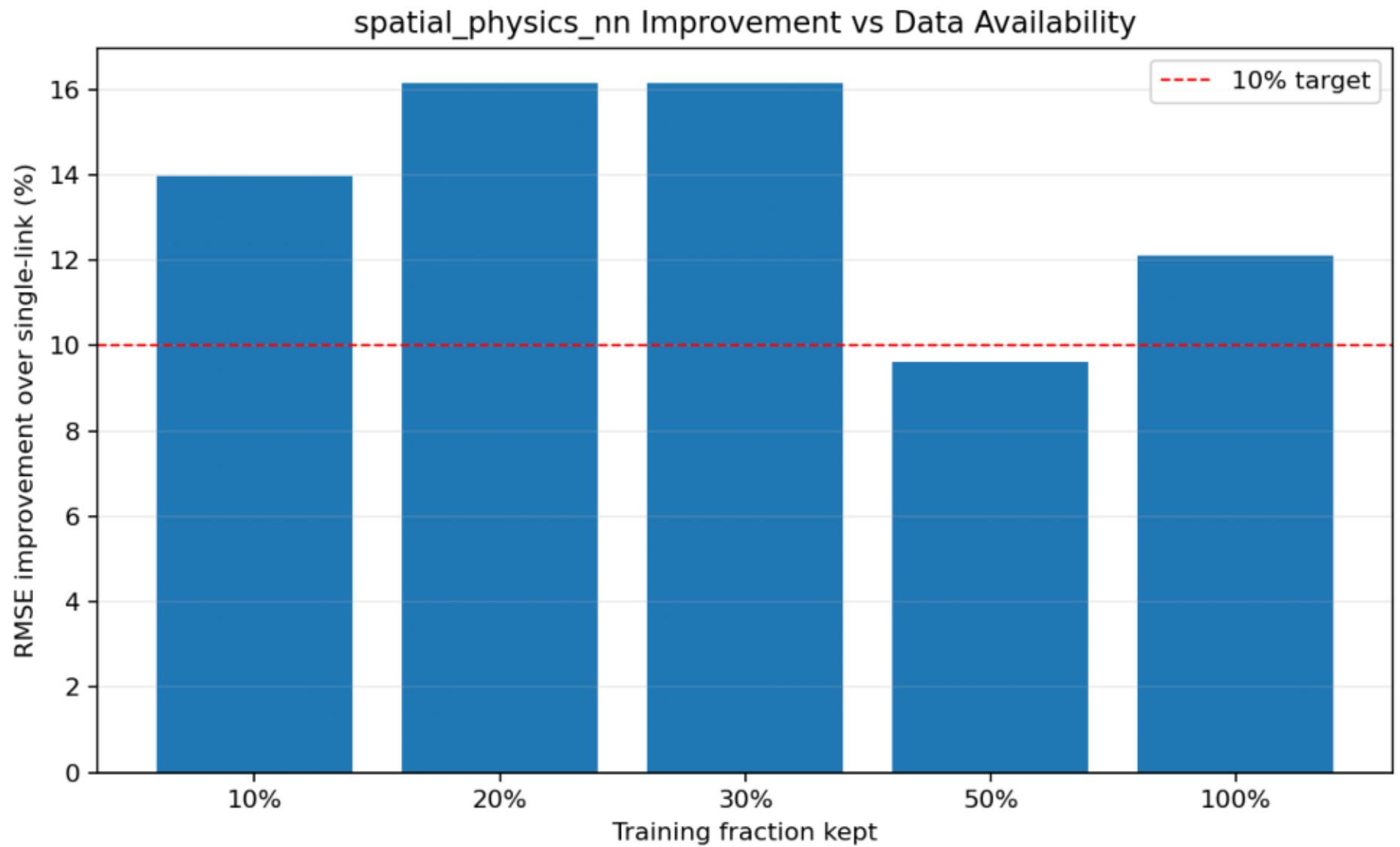
Averaged across seeds 42, 7, 123. Candidate wins 5/5 fractions; 4/5 exceed 10%.

Metric 2 — RMSE Curves Under Sparse Training Data



Fixed validation-selected candidate is compared against the matched single-link baseline.

Metric 2 — Improvement Percentage at Each Data Fraction



Red dashed line = 10% target. Bars above the line meet the robustness target.

Metric 2 — Phase 1 Data Efficiency (Synthetic Oscillator)

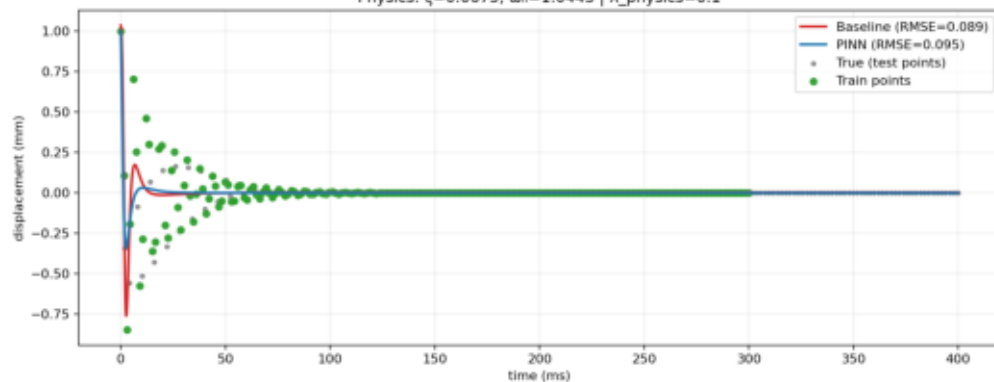
Data	Points	Baseline RMSE	PINN RMSE	Winner
100%	200	0.0692	0.0724	Baseline +5%
50%	100	0.2673	0.1146	PINN +57%
30%	60	0.1506	0.1031	PINN +31%
20%	40	0.2423	0.1070	PINN +56%
10%	20	0.1699	0.1033	PINN +39%
5%	10	0.1619	0.0989	PINN +39%

Crossover between 100% and 50% data. PINN wins below 50%.

Phase 1 — Full Data vs Sparse Data

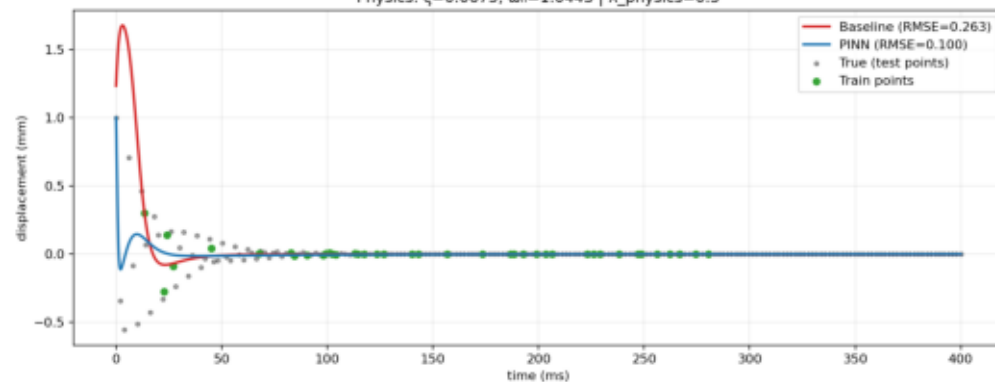
100% data — NN wins

Data Efficiency Test: 100% of training data (200 points)
Training epochs: 500 | no_noise | seed=42
Physics: $\zeta=0.0675$, $\omega_n=1.0443$ | $\lambda_{\text{physics}}=0.1$



20% data — PINN wins

Data Efficiency Test: 20% of training data (40 points)
Training epochs: 1000 | no_noise | seed=42
Physics: $\zeta=0.0675$, $\omega_n=1.0443$ | $\lambda_{\text{physics}}=0.5$



Metric 3: Rainfall Estimation & Classification

PART A – Rain-Rate Estimation (same evidence as Metric 1)

Final test NRMSE/RMSE improvement: 14.5%

STATUS: ACHIEVED

PART B – Wet/Dry Classification (Phase 3E)

Experiment: rain_wetdry.py

Input: outputs/phase2_spatial_validation/prepared_with_splits.csv

Method: Dedicated WetDryMLP classifier with BCEWithLogitsLoss
and positive-class weighting. 4 configurations tested.

Results:

Best classifier: spatial_physics_classifier F1 = 0.720

95% bootstrap F1 CI: [0.47, 0.889]

Single-link baseline F1 = 0.636

Test wet positives = 12

Caveat: the OpenMRG test split is rain-sparse, so the wide
confidence interval should be reported with the headline F1 gain.

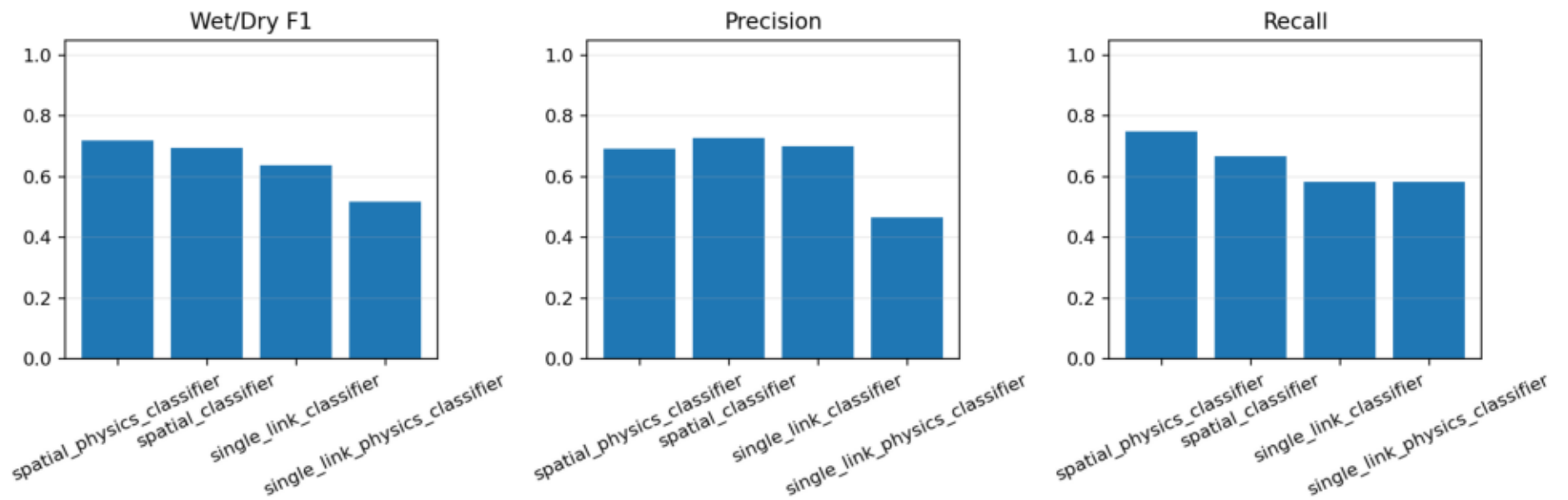
Improvement: 13.1% F1 gain over single-link baseline
(target was >5%)

STATUS: ACHIEVED

Reproduce:

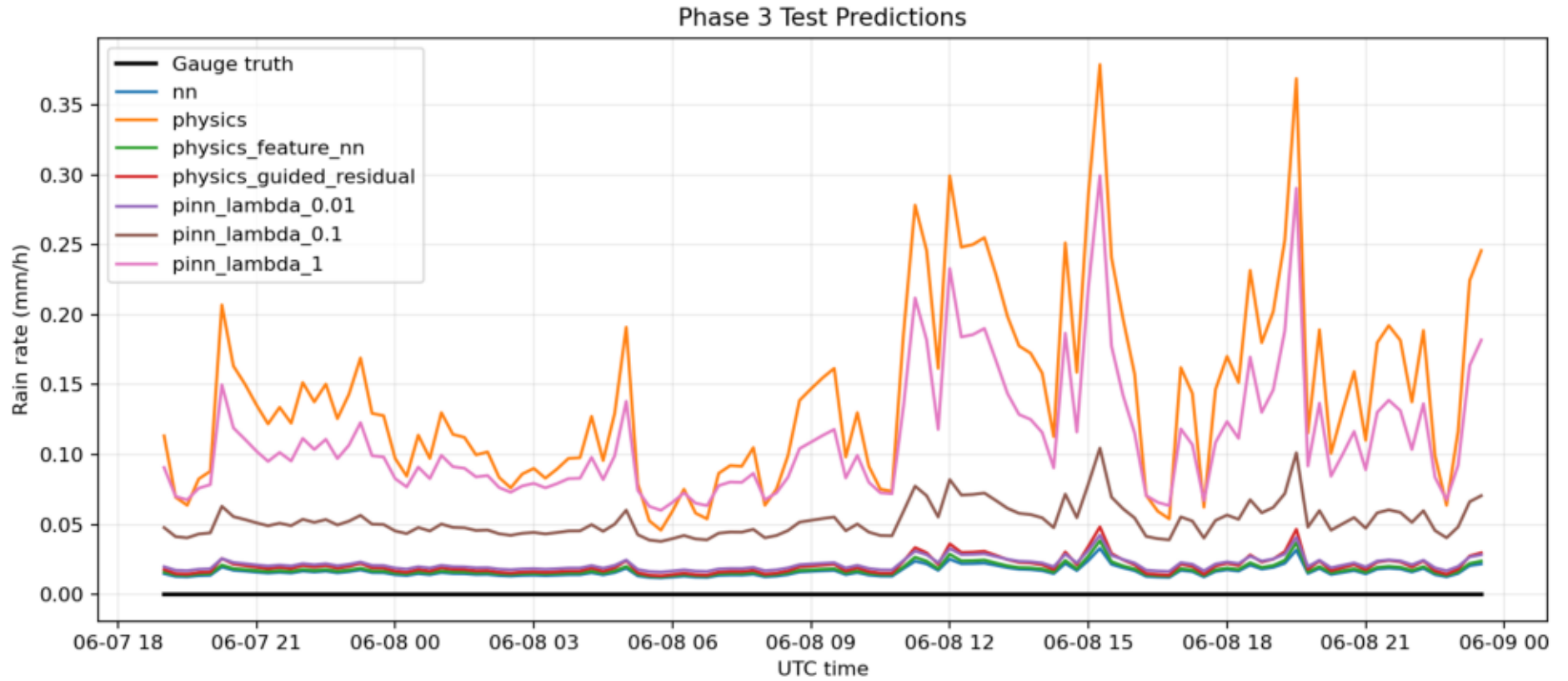
```
python -m src.phase3.rain_wetdry \  
  --input outputs/phase2_spatial_validation/prepared_with_splits.csv \  
  --max-links-per-target 2 --epochs 1000 \  
  --threshold-strategy fixed \  
  --output-dir outputs/phase3e_wetdry
```

Metric 3B — Wet/Dry Classification: F1, Precision, Recall



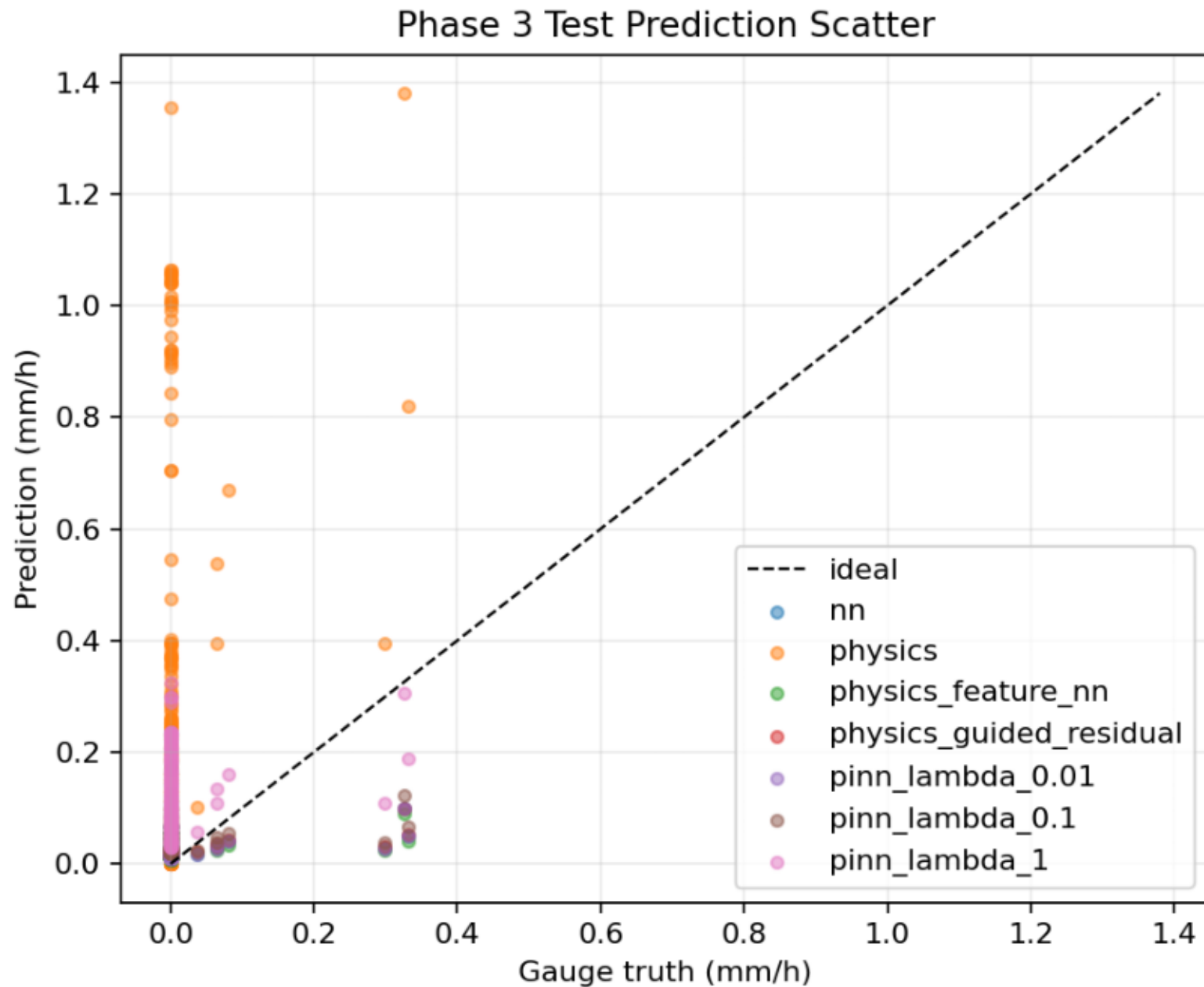
spatial_physics_classifier achieves best F1 (0.720), 13.1% over single-link.

Metric 3A — Phase 3 Test Predictions vs Gauge Truth



Time-series comparison of all Phase 3A/3B models against rain gauge ground truth.

Metric 3A — Predicted vs Observed Rain Scatter



Points near the diagonal indicate accurate predictions.

External Validation: OpenRainER Italy

Purpose: test whether the OpenMRG findings transfer to a separate CML/rain-gauge collection site without using it as the primary claim source.

Dataset: OpenRainER Italy

Rain-rate/robustness subset: January 2021

Prepared table: outputs/prepared/openrainer_202101.csv

Validation report: outputs/phase2_openrainer_202101_validation/

Data quality:

- 11,741 aligned CML/gauge rows

- 4 usable CML sublinks

- test rainy fraction: 11.9%

- test wet rows: 212

Rain-rate estimation:

- single-link RMSE: 1.5425

- spatial+physics RMSE: 1.3424

- improvement: 13.0%

Robustness:

- sparse-data best improvement: 17.2% (5/5 fractions $\geq 10\%$)

- 10% noise best improvement: 17.2% (5/5 fractions $\geq 10\%$)

Wet/dry external check:

- phase3e_openrainer_202104_wetdry_k4_val: spatial+physics $F1=0.617$, baseline $F1=0.563$, improvement=9.5%, 95% CI=[0.563, 0.667], wet positives=149

- Interpretation: OpenRainER strengthens rain-rate and robustness.

- April 2021 adds secondary wet/dry support, but OpenMRG remains the primary wet/dry claim source.

Final Summary — All Deliverable Metrics

Metric	Target	Result	Status
1. RMSE improvement	10-15%	14.5%	ACHIEVED
2. Data efficiency	Maintain with 30-50% less data	16.2% best sparse-data improve	ACHIEVED
2. Noise resilience	>10% RMSE reduction under 10%	11.8% best noise improvement (2/5 >=	PARTIAL
3. Rain estimation	10-20% NRMSE improvement	14.5%	ACHIEVED
3. Wet/dry F1	>5% F1 improvement	13.1%	ACHIEVED

Statuses are computed from the current CSV artifacts.

Key Takeaways

1. Phase 3 rain models are feed-forward MLPs over tabular/spatial CML features. PyNNcml is used for data preparation, not as the trained RNN/LSTM benchmark being extended.
2. Rigid power-law PINN loss (Phase 3A) over-constrains the model and performs WORSE than the pure NN on real CML data.
3. Physics as a soft feature (r_{physics} = power-law rain estimate) and spatial CML context should be reported separately.
Current final-test improvement: 14.5%; Wet/Dry F1 gain: 13.1%.
4. Fixed robustness comparison uses ``spatial_physics_nn``.
Best sparse-data improvement: 16.2%; best 10% noise improvement: 11.3%.
Only 2/5 noisy fractions exceed 10%, so noise support is partial.
5. Phase 1 confirms the classical PINN story on synthetic data (PINN wins below 50% data). Phase 3 applies the same physics motivation to real CML data using a pragmatic MLP physics-feature and spatial-context approach.